

S&P GLOBAL × CRISIL CAMPUS HACKATHON 2026

AI/NLP FINANCIAL RISK ENGINE

From headline to capital impact in seconds

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Live demo: manipal-arsh-agrawal-hackathon.streamlit.app

Code: github.com/agr-arsh-15/manipal-arsh-agrawal-hackathon

Made with **GAMMA**

Risk shows up in text before it shows up in prices

Text

Downgrades, sanctions headlines,
CEO tweets

Manual analysis; capital impact days later

Signals

Sentiment -1 to +1

Event type: 9 classes

Impact: 1-10

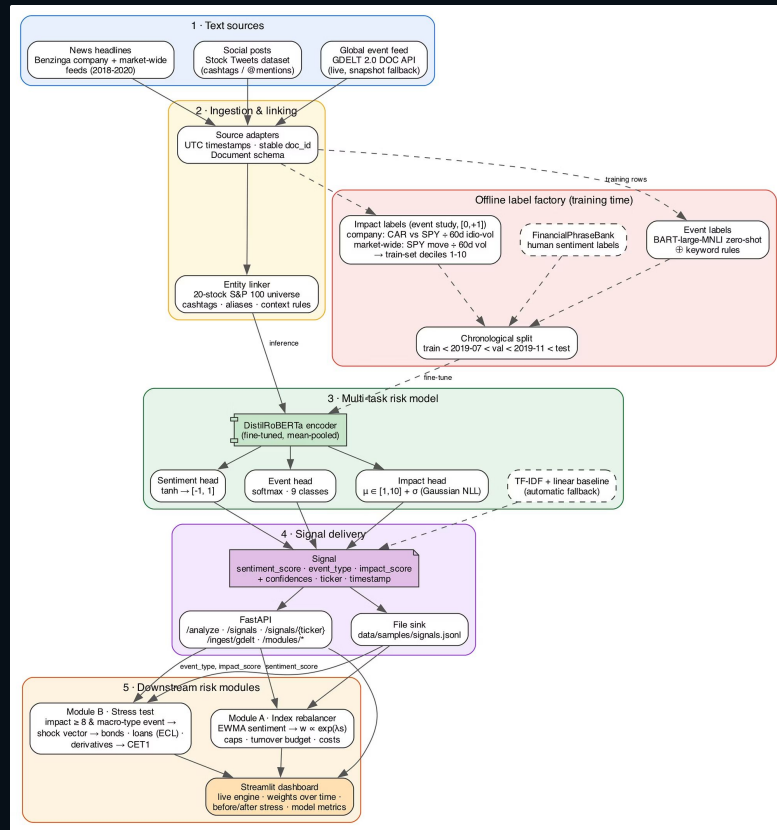
Auditable; anchored to market reaction

Decisions

Module A tilts equities toward
improving sentiment

Module B stress-tests severe
systemic events

One engine, three sources, two downstream modules



1. Sources: financial news, social posts, GDELT live global event feed
2. Ingestion adapters: one common Document schema
3. Entity linker: cashtags and aliases → 20-stock universe
4. Multi-task DistilRoBERTa: sentiment, event, impact + uncertainty
5. FastAPI service + signal store (27,669 signals)
6. Module A (index rebalancer) and Module B (stress testing)
7. Streamlit dashboard (public, live)

Every signal carries source, timestamp, model version, confidence and a text excerpt.

Design choices a risk team can trust

Measured impact

Abnormal returns, headline day + next; stock-volatility scaling; deciles

No look-ahead

Chronological split; after-close → next session; t weights → t+1 returns

Three model heads

DistilRoBERTa; impact uncertainty; 12 ms/headline on CPU

Runs anywhere

Windows/macOS/Linux CI; 64 tests; dashboard smoke test; TF-IDF fallback

Transformer beats the baseline; signals feed real decisions

TRANSFORMER / TF-IDF

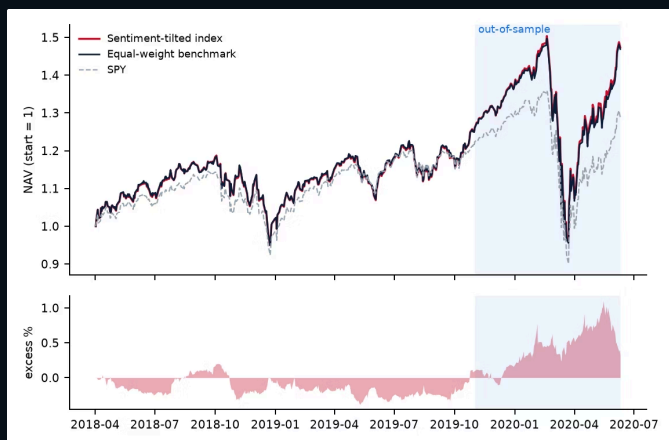
0.78 / 0.40

Event macro-F1

Sentiment r .84/.58 · accuracy .90/.80

High-impact recall .32/.09

MODULE A



Sharpe 0.81 / 0.80 equal weight

Small edge, not statistically significant

MODULE B · SYNTHETIC \$9.1BN BOOK

event_type

Macroeconomic

↑ confidence 68%

impact_score

10

↑ confidence 37%

sentiment_score

-0.31

Trigger gate	Result	This signal	Required
Event type is systemic	Pass	Macroeconomic	Geopolitical, Macroeconomic, Credit Event, Operational/Supply-chain, Regulatory/Legal
Impact is high	Pass	10	>= 8
Sentiment is adverse	Pass	-0.31	< +0.00
Event classification is confident	Pass	68%	>= 60%

Triggered by the model. All trigger gates passed, so the matching scenario ran automatically.

Scenario: Hawkish policy surprise: +200bp rates, equity drawdown, wider credit spreads. Severity 100% → equity -10.0%, rates +200bp, IG +50bp, HY +150bp, USD +3.0%, commodities -5.0%

Portfolio value	Stressed P&L ⓘ	Expected credit loss ⓘ	CET1 ratio ⓘ	Stage 2 loans ⓘ
\$8.5bn	-5.92%	\$57.8m	5.39%	+8
↓ -\$36.1m		↑ +\$34.9m	↓ -7.11 pp	

CET1 falls below the 7% minimum-plus-buffer threshold.

-\$482m

P&L -5.3% · impact 9

6.1%

CET1 from 12.5%; below 7% buffer

Why it matters for a bank or ratings business



Speed
Days → seconds



Credit early
warning

IFRS 9 Stage 2
candidates before
ratings move



Capital planning

CET1 consumption
informs hedging and
limits



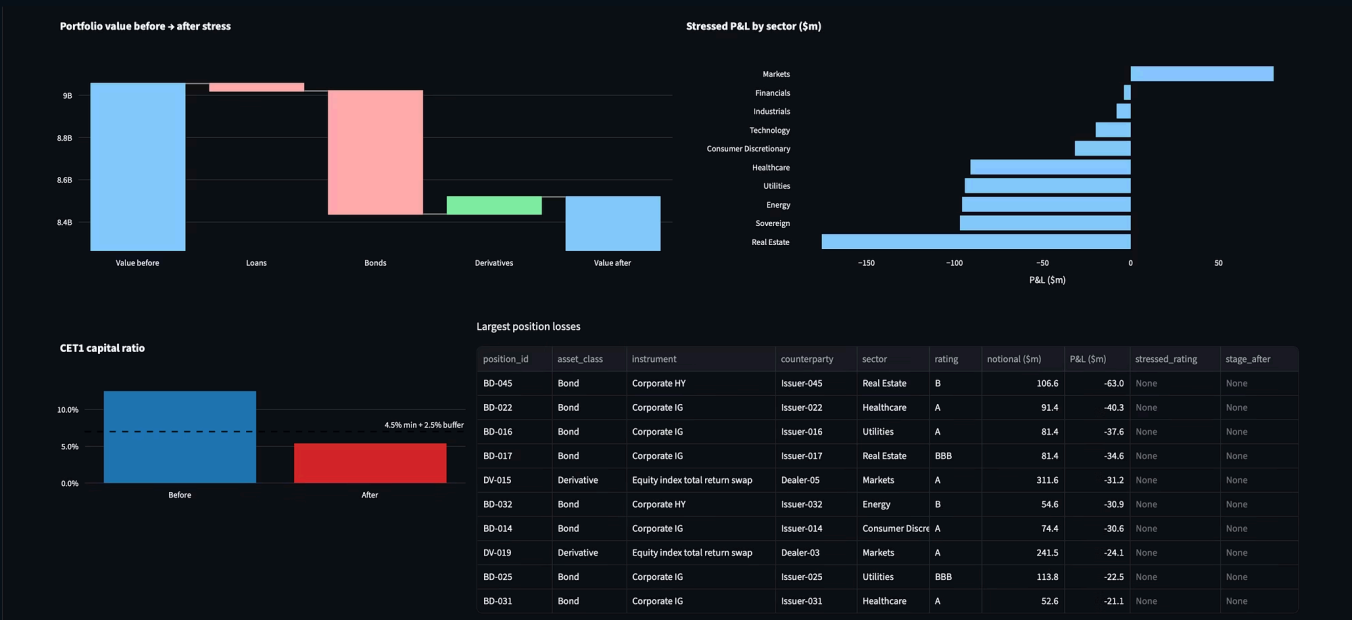
Auditability

Source, version,
confidence

News triage

Prioritise
scenarios

Human review



Honest about what it is, clear on what comes next

LIMITATIONS

- Event labels are weak labels (zero-shot + rules); 268-headline human check in progress
- Headline-level impact is noisy; weak ranking signal for market-wide news
- 2018–mid-2020, 20 large caps; banking book is synthetic
- Module B is a sensitivity model (duration/convexity, delta-gamma), not full revaluation

NEXT STEPS

- Finish the human-labelled evaluation set
- Intraday prices; broader universe and recent data
- Streaming GDELT ingestion with alerting
- Full revaluation and second-round effects

Try it live: manipal-arsh-agrawal-hackathon.streamlit.app